

$c, t \in \mathbb{Z}_2^n, C = U_a |c\rangle_n |t\rangle_n = \begin{cases} |c\rangle_n \cdot |a^c + t \pmod N\rangle_n & 0 \leq t < N \\ |t\rangle_n & N \leq t < 2^n \end{cases}$ or quantum circuit 2 implement

$i \in \mathbb{Z}_n, a_i \equiv a^{2^i} \pmod N \Rightarrow a_i \equiv a_{i-1}^2 \pmod N, C = \sum_{i=0}^{n-1} c_i 2^{n-1-i} \quad (0 \leq i < n, c_i \in \{0, 1\})$ 이면 $0 \leq i \leq n$
 $t < N$ 이면 $t_i = \prod_{l=0}^{i-1} a_{n-l-1}^{c_l} t \pmod N, t \geq N$ 이면 $t_i = t$ 이고 $t < N$ 이 경우: $t_0 = \prod_{l=0}^{-1} a_{n-l-1}^{c_l} t \pmod N = t$, $t_n = \prod_{l=0}^{n-1} a_{n-l-1}^{c_l} t \pmod N = \prod_{l=0}^{n-1} a^{c_l 2^{n-l-1}} \cdot t \pmod N = a^{\sum_{l=0}^{n-1} c_l 2^{n-l-1}} t \pmod N = a^C t \pmod N$ 이다.
 $0 \leq i \leq n-1$ 이면 $t_{i+1} = a_{n-i-1}^{c_i} \cdot t_i \pmod N$ 이다.

$x, y \in \mathbb{Z}_2^n, U_n |x\rangle |y\rangle = \begin{cases} |x\rangle |xy \pmod N\rangle & (x, y) \in \mathbb{Z}_n^2 \cap \text{gcd}(y, N) = 1 \\ |x\rangle |y\rangle & (x, y) \in \mathbb{Z}_n^2 \cup \text{gcd}(y, N) \neq 1 \end{cases}$: bijective and unitary, $O(n^2)$

$U_p: |a\rangle_n \otimes \bigotimes_{i=1}^{n-1} |0\rangle_n \rightarrow \bigotimes_{i=0}^{n-1} |a_i\rangle_n, |y\rangle = |a_0\rangle_n \otimes \bigotimes_{i=1}^{n-1} |y_i\rangle_n, |y_{i-1}\rangle |y_i\rangle = CNOT_n |y_{i-1}\rangle |y_i\rangle,$
 $|y_{i-1}\rangle |y_i\rangle \leftarrow U_n |y_{i-1}\rangle |y_i\rangle : \text{for } i=1 \sim n-1$ 이면 $U_p(y) = |a_0\rangle \cdots |a_{n-1}\rangle, O(n^3)$

P.F) $U_n(CNOT_n(|a\rangle_n |0\rangle_n)) = U_n(|a_0\rangle_n |a_0\rangle_n) = |0_0\rangle_n |a_0^2 \pmod N\rangle_n = |a_0\rangle_n |a_1\rangle_n (a_0 \equiv a \pmod N = a)$
 그리고 $\bigotimes_{i=1}^{n-1} U_n(CNOT_n(|a_i\rangle_n |0\rangle_n)) \cdots = |a_0\rangle_n \cdots |a_{i-1}\rangle_n U_n(|a_i\rangle_n |a_i\rangle_n) = |a_0\rangle_n \cdots |a_{i-1}\rangle_n$

$\text{귀납법} \Rightarrow \sum |a_0\rangle_n \dots |a_{i-1}\rangle_n U_m (NOT_n (|a_i\rangle_n |0\rangle_n) \dots = |a_0\rangle_n \dots |a_{i-1}\rangle_n U_m (|a_i\rangle_n |a_i\rangle_n) = |a_0\rangle_n \dots |a_{i-1}\rangle_n$
 $|a_i\rangle_n |a_i \bmod N\rangle_n |0\rangle_n \dots = |a_0\rangle_n \dots |a_{i+1}\rangle_n |0\rangle_n \dots \quad \text{이므로} \quad \text{성립} \quad \text{loop } (U_m: O(n^2)) \therefore O(n^3)$

$(-U_a: |\psi\rangle = |c\rangle_n |a\rangle_n |0\rangle_n^{\otimes n-1} |1\rangle_n |t\rangle_n, U_p \sum |\psi\rangle = |c\rangle_n |a_0\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t\rangle_n, i = 1 \sim n$

$|\psi\rangle = \text{SWAP}_{i-1} |\psi\rangle, |\psi_{n+1}\rangle |\psi_{n+2}\rangle = U_m |\psi_{n+1}\rangle |\psi_{n+2}\rangle, |\psi\rangle = \text{SWAP}_{i-1} |\psi\rangle$

$P \in) (\text{SWAP}_i |\psi\rangle = |c\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t\rangle_n \text{ if } c_0 = 0, |c\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |a_{n-1}\rangle_n |t\rangle_n \text{ if } c_0 = 1$

$\therefore U_m |1\rangle_n |t\rangle_n = |1\rangle_n |t\rangle_n = |1\rangle_n |a_{n-1}^{c_0} \bmod N\rangle_n = |1\rangle_n |t_1\rangle_n, U_m |a_{n-1}\rangle_n |t\rangle_n = |a_{n-1}\rangle_n |t\rangle_n \text{ or}$

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$|a_{n-1}\rangle_n |a_{n-1}^{c_0} \bmod N\rangle_n = |a_{n-1}\rangle_n |t_1\rangle_n, \text{즉} (\text{SWAP}_i: \psi = |c\rangle_n |a_0\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t_1\rangle_n \text{ 이다.}$

$\text{귀납법} \Rightarrow i = n: |\psi\rangle = |c\rangle_n |a_0\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t_n\rangle_n \therefore |c\rangle_n |t_n\rangle_n = (-U_a |c\rangle_n |t\rangle_n$

$\text{Oscilla bit: } O(n^2), U_p: O(n^2), \text{loop } (\text{SWAP}_i: O(1), U_m: O(n^2)) \therefore O(n^3) \text{ 이다.}$

$$f: \mathbb{N}^2 \rightarrow \mathbb{N} \text{ with } \bar{w}_f = 4 \quad (6.4.6) \quad n = O(\log N) \therefore O(n^3) = O((\log N)^3)$$

integer factorization problem: classical, fully analyzed Monte Carlo algorithm

$$: e^{(1+o(1))(\log N \cdot (\log \log N)^{\frac{1}{2}})}$$

$$\text{heuristic Monte Carlo algorithm: } e^{(\frac{4}{3} + o(1)) \cdot (\log N)^{\frac{1}{3}} \cdot (\log(\log N))^{\frac{2}{3}}}$$

$a \in \mathbb{Z}_n$: select, $d = \gcd(a, N)$ compute, $d=1 \Rightarrow$ Find order $(N, a, 2\lceil \log N \rceil + 1) \in \mathbb{Z}$

$$a^r \equiv 1 \pmod{N} \text{ if } r \text{ is a multiple of } \frac{N-1}{2} \Rightarrow \text{if } r \text{ is even } (a^{r/2-1} / a^{r/2+1}) \equiv O(\log N) \wedge a^{r/2+1} \Rightarrow \gcd(a^{r/2+1}, N):$$

$: N$ is proper divisor with probability at least $O(1/q, O((\log N)^3)$

P : odd prime number, $e \in \mathbb{N}$, $P = 2^d m + 1$, $2 \nmid m$, $a \in \mathbb{Z}_P^*$, $a^r \equiv 1 \pmod{P^e}$, $2^d | r \text{ and } \frac{r}{2^d} \equiv 1/2$

$P \nmid 1$ $\varphi(P^e) = (P-1)P^{e-1}$, $g \in \mathbb{Z}_P^*$, $g^{\varphi(P^e)} \equiv 1 \pmod{P^e}$ and $\mathbb{Z}_P^* \rightarrow \mathbb{Z}_P^*$, $\kappa \mapsto g^\kappa \pmod{P^e}$: bijective,

$a = g^\kappa \pmod{P^e} \Rightarrow a$ is order $r = \frac{\varphi}{\gcd(\kappa, \varphi)}$ and $\varphi \in \mathbb{N}$: even $\frac{1}{2}$, odd $\frac{1}{2} \therefore 2^d | r : 1/2$

$$N = \prod_{i=1}^n p_i^{e_i}, a \in \mathbb{Z}_N^*, a \bmod N \text{ of order } r: \exists (r, a^{r/2} \not\equiv \pm 1 \pmod{N}) \Rightarrow 1 \sim 2^{1-m}$$

$$P \in \{ a_i \in \mathbb{Z}_{p_i^{e_i}}, a \in \mathbb{Z}_N^*, a \equiv a_i \pmod{p_i^{e_i}}, r: \text{order of } a \bmod N, r_i: \text{order of } a_i \bmod p_i^{e_i}$$

$$r = \text{lcm}(r_1, \dots, r_n), r = 2^f m, r_i = 2^{f_i} m_i, (m, m_i: \text{odd}), r: \text{odd or } r: \text{even}, a^{r/2} \equiv -1 \pmod{N} \Rightarrow 1 \leq i \leq m, f = \tau.$$

$$r: \text{odd}: 2 \nmid r, \exists (r, a^{r/2} \equiv -1 \pmod{N}) \Rightarrow a^f \equiv 1 \pmod{p_i^{e_i}}, a^{r/2} \equiv -1 \pmod{p_i^{e_i}}: \therefore r, r_i \nmid r/2: \therefore f = f_i$$

$$\therefore r: \text{odd or } \exists (r, a^{r/2} \equiv -1 \pmod{N}) \leq \forall i, 1 \leq i \leq m, f_i = f_j = \frac{1}{2^{m-1}} \therefore$$

$$1 - \frac{1}{2^{m-1}} \geq \frac{1}{2} \therefore 0.999 \cdot \frac{1}{2} \approx 0.199, O(\log N |^3)$$

$$2 \nmid N, a, b \in \mathbb{Z}_N^*, r: \text{order of } a \bmod N, a^t \equiv b \pmod{N} : \tau: \text{discrete logarithm of } b \text{ to base } a \bmod N, \text{ we may assume } r \text{ is prime by the Pohlig-Hellman algorithm, } t > 1$$

$$n \in \mathbb{N}, U_a, U_b: |x\rangle_n \rightarrow \begin{matrix} |u \text{ or } b\rangle_n \\ |x\rangle_n \end{matrix} \xrightarrow{U_a} \begin{matrix} |u \text{ or } b\rangle_n \\ |u \text{ or } b\rangle_n \end{matrix} \xrightarrow{U_b} \begin{matrix} |u \text{ or } b\rangle_n \\ |u \text{ or } b\rangle_n \end{matrix}, U_a \text{ is eigenvalue: } a^{2\pi i \cdot \frac{t}{r}}, U_b \text{ is eigenvalue: } a^{2\pi i \cdot \frac{t}{r}}$$

$$0 \leq k < r, \text{ by quantum phase estimation, } (x, y) \in \mathbb{Z}_m \times \mathbb{Z}_r, K = \lfloor \frac{r}{2^n} \rfloor, K \neq \bmod r = \lfloor \frac{r}{2^n} \rfloor: \therefore t = K^2 \lfloor \frac{r}{2^n} \rfloor \bmod r$$

$n = \lceil \log_2 r \rceil + 1$: probability $\frac{64(r-1)}{r^4}$, $O((\log(V))^3)$

$P(t) |\psi_0\rangle = |0\rangle_n |0\rangle_n |1\rangle_n$, $|\psi_1\rangle = \left(\frac{107+117}{\sqrt{2}}\right)^{\otimes n} \left(\frac{107+117}{\sqrt{2}}\right)^{\otimes n} |1\rangle_n$, $(-U_a \text{ operator } |\psi_{2a}\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} (|\psi_k(\frac{a}{r})\rangle$
 $\cdot (\sum_{k=0}^{r-1} |\psi_k\rangle) \cdot |u_n\rangle)$ $(-U_b \text{ operator: } |\psi_{2b}\rangle = \frac{1}{r} \sum_{k=0}^{r-1} (|\psi_k(\frac{a}{r})\rangle |\psi_k(\frac{b}{r})\rangle |u_k\rangle)$, $|t_3\rangle = (\frac{1}{r}, |\psi_k(\frac{a}{r})\rangle, |\psi_k(\frac{b}{r})\rangle)_{0 \leq k < r}$

apply QFT_n^{-1} : $|\psi_4\rangle = (\frac{1}{r}, QFT_n^{-1} |\psi_k(\frac{a}{r})\rangle QFT_n^{-1} |\psi_k(\frac{b}{r})\rangle)_{0 \leq k < r}$ probability $\frac{64}{r^4}$, $(x,y) \in \mathbb{Z}_m \times \mathbb{Z}_n$,
 $|\Delta(\frac{a}{r}, n)\rangle | \leq \frac{1}{2n} < \frac{1}{2r}$, $|\Delta(\frac{kt_{mod}r}{r}, n, y)\rangle | \leq \frac{1}{2n} < \frac{1}{2r}$ $0 \leq z$ $K = \lfloor \frac{r^n}{2n} \rfloor$, $kt_{mod}r = \lfloor \frac{r^n}{2n} \rfloor$

Classical public-key cryptography is vulnerable - Quantum-resistant cryptography

Group G , $H < G$, set X , $f: G \rightarrow X$, $\forall g, g' \in G$, $gH = g'H \Leftrightarrow f(g) = f(g')$ $0 \leq i$ f : hides subgroup H

$\therefore f$ has the same value for all elements of a coset of H ,

input: f , output: finite generating system for H

Deutsch: $f: \{0,1\} \rightarrow \{0,1\}$, f is constant: f hides $\{0,1\}$, f is balanced: f hides $\{0\}$

Simon's problem: $f: \{0,1\}^n \rightarrow \{0,1\}^n$, $\vec{z} \in \{0,1\}^n$, $\vec{z} \neq \vec{0}$, $\forall \vec{x}, \vec{y} \in \{0,1\}^n$, $\vec{y} = \vec{x} \oplus \vec{z} \Leftrightarrow f(\vec{x}) = f(\vec{y})$
 $H = \{\vec{0}, \vec{z}\}$; Generating system $(\vec{z})$, $(\vec{0}, \vec{z})$

$N \geq 3$, $2 \nmid N$, $a \in \mathbb{Z}_N$, $\gcd(a, N) = 1$. $f: (\mathbb{Z}_N, +) \rightarrow \mathbb{Z}_N$, $f(i) = a^i \pmod{N}$, $H = r\mathbb{Z} : (r)$

$a, b \in \mathbb{Z}_N^*$, r : order of $a \pmod{N}$. $f: (\mathbb{Z}_r \times \mathbb{Z}_r, +) \rightarrow \mathbb{Z}_N$, $(x, y) \mapsto a^x b^y$, $H = \mathbb{Z}_r(1, -t)$

r, t, m s.t. $H = \mathbb{Z}_r(1, -t) \subseteq$ generating system $((x_0, y_0), \dots, (x_{m-1}, y_{m-1}))$, $d = \gcd(x_0, \dots, x_{m-1})$

$\gcd(d, r) = 1$. $t = d^{-1} \gcd(y_0, \dots, y_{m-1}) \pmod{r}$

$P \in \mathbb{Z} \exists u_0, \dots, u_{m-1} \in \mathbb{Z}$, $\sum_{i=0}^{m-1} u_i (x_i, y_i) \equiv (1, -t) \pmod{r}$, $(x_i, y_i) \in \mathbb{Z}_r(1, -t)$, $\sum_{i=0}^{m-1} u_i x_i (1, -t) \equiv (1, -t) \pmod{r}$

$\therefore x'_i = x_i / d \pmod{r}$, $d \sum_{i=0}^{m-1} u_i x'_i \equiv 1 \pmod{r} \because \gcd(d, r) = 1$, $\gcd(y_0, \dots, y_{m-1}) \equiv td \pmod{r}$